A Structural Theory of the Willingness-to-Pay Gap: Valuation and Behaviour in Climate Mitigation

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Part I: Research Question

Research Question: If people value climate change highly in a survey, why are they under-contributing?

- ➢ Environmental economics often compares stated WTP and actual contribution as if they were two measurements of the **same underlying WTP**.
- ➢ We claim that they are **not necessarily the same object**
- We compare two coexisting frameworks under the same preference structure:
  i. WTP as Compensating Variation (**Valuation Model**)
  ii. WTP as Optimisation Behaviour (**Choice Model**)
- We show that a **wedge arises generically**.

Part II: Model Construction

Primitives and Assumptions

- $U_{i,t}(c, W, E, M) = u_i(c - W_t) + E_{i,t} + M_{i,t}$.

  where c - private consumption, W - WTP, E - environmental improvement,

  M - moral satisfaction.

- **A1 (Regularity)** *The general utility function is additively separable and twice continuously differentiable.*

- **A2 (Monotonicity and concavity)** *All component utility* functions *are monotonic and concave. In particular,*

$$u'(c - W_i) > 0, \qquad u''(c - W_i) < 0,$$

$$E'(p_i Q) \geq 0, \qquad E''(p_i Q) \leq 0,$$

$$M'(W_i) \geq 0, \qquad M''(W_i) \leq 0.$$

1. WTP as Compensating Variation

- WTP is defined implicitly by:

  $$\{u_i(c - W^{CV}) + E_{i,1}(\bar{W}) + M_{i,1}(\bar{W})\} = \{u_i(c) + E_{i,0}(\bar{W}) + M_{i,0}(\bar{W})\},$$

  where $\bar{W}$ is the voluntary equilibrium WTP

- Interpretations:
  - WTP is an **implied welfare measure**; it is **not chosen**
  - The **partial equilibrium** (use of $\bar{W}$) captures the **hypothetical environment** in the survey

Discusses at the end

2. WTP as Optimisation

- Now WTP is a choice variable, subject to

$$W_i^* \in \arg\max_{W_i \geq 0} \{u(c - W_i) + E_i(W_i) + M_i(W_i)\}$$

- Individuals **choose** their WTP level, W_i^*
- WTP solves the classic problem: **MB=MC**
- We will assume that an interior solution W^* exists

Part III: The Wedge $\Delta = W^{CV} - W^*$

The Wedge Δ

- Define:

$$\Delta = W^{CV} - W^{*}$$

- **Proposition 1 (Existence of the wedge)** *Under A1 and A2, and assuming an interior solution exists,* W_i^* *and* W_i^{CV} *are generally distinct, so* $\Delta \neq 0$ *except for knife-edge conditions, when the level condition and the first-order condition coincide.*

- Why?
  - ➢Compensating variation evaluates **welfare changes** between states, while behavioural WTP is determined by **marginal incentives**.
  - ➢Coincidence occurs when the compensating-variation payment satisfies the behavioural first-order condition.

Sign and Mechanism

- **Lemma 2 (Sign of the wedge)** *Under A1 and A2, for any* $w \in [0, c]$,

$$w \lesseqgtr W_i^*(s) \quad \Leftrightarrow \quad U_W(w; s) \gtreqless 0.$$

Since we define the wedge Δ *as* $\Delta = W^{CV}(s) - W^*(s)$. *Therefore,*

$$\Delta \lesseqgtr 0 \Leftrightarrow U_W(W^{CV}(s); s) \gtreqless 0.$$

- Under Lemma 2, we can evaluate the sign of the wedge without explicitly solving for W_i^*.
- Under the general form, the sign of the wedge is **ambiguous**.
- If marginal utility at the stated-WTP benchmark is negative, the actual contribution lies below stated WTP.

Real-World Implication

- **Remark 3 (Sign of the wedge under weak marginal incentives)**
  Suppose at a low level of observability ($s \to 0$),
  $$U_W(W^{CV}; s) < 0.$$
  Then, under Lemma 2,
  $$\Delta = W^{CV} - W^* > 0.$$
- i.e. if observability is sufficiently low → reputational return is weak → marginal benefit is small → real contribution is less than stated.
- Low-observability environments($s \to 0$), such as the anonymous donation, correspond to this case.

Observability and a Numerical Illustration

Proposition 2 (Sensitivity condition) Since $\Delta = W_i^{CV} - W_i^*$, we have

$$\frac{d\Delta}{ds} < 0 \iff \frac{dW^*}{ds} > \frac{dW_i^{CV}}{ds}.$$

s	$W^{CV}(s)$	$W^*(s)$	**Wedge** $\Delta_i = W_i^{CV}(s) - W_i^*(s)$
0.0	90.91	66.48	+24.42
0.2	90.91	95.47	-4.56
0.5	90.91	98.08	-7.17
0.8	90.91	98.78	-7.87
1.0	90.91	99.02	-8.11

Table 1: The Parametric Example Results

Part IV: Summary & Discussion

Summary & Implication

- WTP is not one concept, but
  - Under CV, it's the **welfare value** from changing states.
  - Under optimisation, it's the behavioural contribution chosen under **marginal incentives**.
    - ➢They are conceptually distinct objects.
- Policies that enhance observability—such as public recognition or visible commitments—can potentially reduce the gap between stated WTP and actual behaviour.

❖Observability is one mechanism within our model construction, not the only factor that could shift the wedge.

Literature, Extensions & Discussions

- CV-WTP is a welfare-level valuation object (Hicks, 1939), and stated-preference design has been used to recover this welfare (Carson and Groves, 2007; etc.).
  - ➢**Survey answers should be expected to differ from behaviour.**

- Dynamic settings may depend on perceived equilibrium **(**Zhao and Kling, 2004), motivating the extension of **dynamic learning/equilibrium**.

- This paper can be extended to provide a **theoretical foundation for cheap-talk design** in contingent valuation.
  - ➢Standard cheap-talk is usually viewed as **reducing hypothetical bias** (Cummings and Taylor, 1999).
  - ➢My interpretation: cheap talk may **alter the relationship** between stated valuation and behavioural incentives.

More broadly…

- The paper reflects a wider theoretical interest:
  - ➢Behavioural tools (such as cheap talk) are often treated as practical fixes
  - ➢Yet some remain theoretically under-modelled.
- This paper starts from a behavioural puzzle and asks what economic object was being changed, rather than following the *'path of least theory'* (Spiegler, 2024).
- Theory asks: what object changes, through which incentive channel, and why does behaviour change?
  - ➢Economics is not only about measuring effects but also about explaining mechanisms.